



Nagar Yuwak Shikshan Sanstha's  
**Yeshwantrao Chavan College of Engineering**  
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)  
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## Department of Artificial Intelligence and Data Science

### Vision of the Department

*To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.*

### Mission of the Department

*To nurture computer engineering professionals by offering:*

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

## Session 2026-2027

|                                         |                                         |
|-----------------------------------------|-----------------------------------------|
| <b>Vision:</b> Dream of where you want. | <b>Mission:</b> Means to achieve Vision |
|-----------------------------------------|-----------------------------------------|

**Program Educational Objectives of the program (PEO):** (broad statements that describe the professional and career accomplishments)

|      |                             |                                               |                                                                          |
|------|-----------------------------|-----------------------------------------------|--------------------------------------------------------------------------|
| PEO1 | <b>Preparation</b>          | <b>P: Preparation</b>                         | <b>Pep-CL abbreviation<br/>pronounce as Pep-si-IL easy<br/>to recall</b> |
| PEO2 | <b>Core Competence</b>      | <b>E: Environment (Learning Environment)</b>  |                                                                          |
| PEO3 | <b>Breadth</b>              | <b>P: Professionalism</b>                     |                                                                          |
| PEO4 | <b>Professionalism</b>      | <b>C: Core Competence</b>                     |                                                                          |
| PEO5 | <b>Learning Environment</b> | <b>L: Breadth (Learning in diverse areas)</b> |                                                                          |

**Program Outcomes (PO):** (statements that describe what a student should be able to do and know by the end of a program)

### Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

**PSO Keywords:** Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

**Integrity:** I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

**Krishna Mishra**

(Signature in Handwritten)



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|                 |                       |                        |                                    |
|-----------------|-----------------------|------------------------|------------------------------------|
| <b>Session</b>  | <b>2026-27 (Odd)</b>  | <b>Course Name</b>     | <b>LAB: Optimization technique</b> |
| <b>Semester</b> | <b>7<sup>th</sup></b> | <b>Course Code</b>     | <b>23ADS1708</b>                   |
| <b>Roll No</b>  | <b>41</b>             | <b>Name of Student</b> | <b>Krishna Mishra</b>              |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Practical Number   | <b>7</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Course Outcome     | CO1: Understand unconstrained and constrained optimization techniques.<br>CO2: Apply non-linear optimization techniques to solve engineering problems.<br>CO3: Analyze optimization problems using operations research techniques.<br>CO4: Evaluate transportation and transshipment optimization problems.                                                                                                                                                                                                                                                                                                                                                |
| Aim                | To determine the best-fit curve using the Nonlinear Least Squares Method.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Problem Definition | To fit the exponential model $y = a \cdot e^{bx}$ to the given data points ( $x = 0, 1, 2, 3, 4, 5$ and $y = 2.0, 3.4, 5.3, 9.1, 14.5, 24.6$ ) by minimizing the sum of squared residuals, and to determine the parameters $a$ and $b$ of the best-fit curve.                                                                                                                                                                                                                                                                                                                                                                                              |
| Theory (100 words) | The Nonlinear Least Squares method fits a model $y = f(x; \beta)$ that is nonlinear in its parameters $\beta$ to observed data $(x_i, y_i)$ . The best parameters minimize the sum of squared residuals $S(\beta) = \sum [y_i - f(x_i; \beta)]^2$ . Unlike linear least squares, there is no closed-form solution, so iteration is required. The Gauss-Newton method linearizes the model using its Jacobian $J$ and updates $\Delta\beta = (J^T J)^{-1} J^T r$ , where $r$ is the residual vector. The Levenberg-Marquardt algorithm, used by SciPy's <code>curve_fit</code> , adds a damping term for stability. The fit is judged using SSE and $R^2$ . |
|                    | <b>Step-by-Step Method</b><br>1. Select the model $y = a \cdot e^{bx}$ and enter the observed data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |



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|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <ol style="list-style-type: none"><li>2. Define the residual of each point as the observed <math>y</math> minus the model value <math>a \cdot e^{bx}</math>.</li><li>3. Define the objective <math>S(a, b) = \sum r_i^2</math>.</li><li>4. Choose an initial guess for the parameters <math>a</math> and <math>b</math>.</li><li>5. Compute the Jacobian of the model with respect to <math>a</math> and <math>b</math> at the current guess.</li><li>6. Update the parameters using the Gauss-Newton step (with damping in Levenberg–Marquardt).</li><li>7. Repeat until the sum of squared residuals stops decreasing.</li><li>8. Calculate SSE and <math>R^2</math> to check the quality of the fit.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Procedure and Execution (100 Words) | <p>Algorithm:</p> <ol style="list-style-type: none"><li>1. Enter the data points <math>(x, y)</math>.</li><li>2. Define the nonlinear model <math>y = a \cdot e^{bx}</math>.</li><li>3. Provide an initial guess for <math>a</math> and <math>b</math>.</li><li>4. Fit the model using nonlinear least squares (curve_fit).</li><li>5. Obtain the estimated parameters <math>a</math> and <math>b</math>.</li><li>6. Calculate the fitted values and residuals.</li><li>7. Calculate SSE and <math>R^2</math>.</li><li>8. Display the parameters and the best-fit curve.</li></ol> <p>Code:</p> <pre>import numpy as np from scipy.optimize import curve_fit  # Observed data x = np.array([0, 1, 2, 3, 4, 5], dtype=float) y = np.array([2.0, 3.4, 5.3, 9.1, 14.5, 24.6])  # Model to be fitted: y = a * exp(b * x) def model(x, a, b):     return a * np.exp(b * x)  # Nonlinear least squares fit (initial guess: a=1, b=0.1) params, cov = curve_fit(model, x, y, p0=[1.0, 0.1]) a, b = params  # Fitted values and residuals y_fit = model(x, a, b) residuals = y - y_fit  # Error metrics sse = np.sum(residuals**2) sst = np.sum((y - np.mean(y))**2) r2 = 1 - sse/sst  # Sum of Squared Errors # Total Sum of Squares # R-squared</pre> |



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|                                                                       | <pre># Output results print("Model: y = a * exp(b * x)") print("\n x      y(actual)      y(fitted)      residual") for xi, yi, fi, ri in zip(x, y, y_fit, residuals):     print(f"{xi:2.0f}      {yi:8.3f}      {fi:8.3f}      {ri:8.3f}")  print("\nEstimated Parameters:") print(f"a = {a:.4f}") print(f"b = {b:.4f}")  print("\nSum of Squared Errors (SSE) = %.4f" % sse) print("R-squared = %.4f" % r2) print("\nBest-fit curve: y = %.4f * exp(%.4f * x)" % (a, b))</pre>                                                                                                                                                                                                                                      |           |           |           |          |   |       |       |       |   |       |       |       |   |       |       |        |   |       |       |       |   |        |        |        |   |        |        |       |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|----------|---|-------|-------|-------|---|-------|-------|-------|---|-------|-------|--------|---|-------|-------|-------|---|--------|--------|--------|---|--------|--------|-------|
|                                                                       | <p><b>Output:</b></p> <p>Model: y = a * exp(b * x)</p> <table><tr><th>x</th><th>y(actual)</th><th>y(fitted)</th><th>residual</th></tr><tr><td>0</td><td>2.000</td><td>1.960</td><td>0.040</td></tr><tr><td>1</td><td>3.400</td><td>3.249</td><td>0.151</td></tr><tr><td>2</td><td>5.300</td><td>5.384</td><td>-0.084</td></tr><tr><td>3</td><td>9.100</td><td>8.921</td><td>0.179</td></tr><tr><td>4</td><td>14.500</td><td>14.784</td><td>-0.284</td></tr><tr><td>5</td><td>24.600</td><td>24.499</td><td>0.101</td></tr></table> <p>Estimated Parameters:<br/>a = 1.9604<br/>b = 0.5051</p> <p>Sum of Squared Errors (SSE) = 0.1542<br/>R-squared = 0.9996</p> <p>Best-fit curve: y = 1.9604 * exp(0.5051 * x)</p> | x         | y(actual) | y(fitted) | residual | 0 | 2.000 | 1.960 | 0.040 | 1 | 3.400 | 3.249 | 0.151 | 2 | 5.300 | 5.384 | -0.084 | 3 | 9.100 | 8.921 | 0.179 | 4 | 14.500 | 14.784 | -0.284 | 5 | 24.600 | 24.499 | 0.101 |
| x                                                                     | y(actual)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | y(fitted) | residual  |           |          |   |       |       |       |   |       |       |       |   |       |       |        |   |       |       |       |   |        |        |        |   |        |        |       |
| 0                                                                     | 2.000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1.960     | 0.040     |           |          |   |       |       |       |   |       |       |       |   |       |       |        |   |       |       |       |   |        |        |        |   |        |        |       |
| 1                                                                     | 3.400                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3.249     | 0.151     |           |          |   |       |       |       |   |       |       |       |   |       |       |        |   |       |       |       |   |        |        |        |   |        |        |       |
| 2                                                                     | 5.300                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 5.384     | -0.084    |           |          |   |       |       |       |   |       |       |       |   |       |       |        |   |       |       |       |   |        |        |        |   |        |        |       |
| 3                                                                     | 9.100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 8.921     | 0.179     |           |          |   |       |       |       |   |       |       |       |   |       |       |        |   |       |       |       |   |        |        |        |   |        |        |       |
| 4                                                                     | 14.500                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 14.784    | -0.284    |           |          |   |       |       |       |   |       |       |       |   |       |       |        |   |       |       |       |   |        |        |        |   |        |        |       |
| 5                                                                     | 24.600                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 24.499    | 0.101     |           |          |   |       |       |       |   |       |       |       |   |       |       |        |   |       |       |       |   |        |        |        |   |        |        |       |
| Output Analysis                                                       | <p>The program fitted the model <math>y = a \cdot e^{bx}</math> to the data and obtained <math>a = 1.9604</math> and <math>b = 0.5051</math>. The residuals are small, with a maximum absolute value of 0.284. The sum of squared errors is 0.1542 and <math>R^2 = 0.9996</math>, which shows that the exponential curve describes the data very well. The positive value of b indicates exponential growth.</p>                                                                                                                                                                                                                                                                                                     |           |           |           |          |   |       |       |       |   |       |       |       |   |       |       |        |   |       |       |       |   |        |        |        |   |        |        |       |
| Link of student Github profile where lab assignment has been uploaded | <a href="https://github.com/krish-na45/Optimization-Technique-Practicals">https://github.com/krish-na45/Optimization-Technique-Practicals</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |           |           |          |   |       |       |       |   |       |       |       |   |       |       |        |   |       |       |       |   |        |        |        |   |        |        |       |
| Conclusion                                                            | <p>The best-fit curve was successfully determined using the <b>Nonlinear Least Squares Method</b>. The best-fit curve is <math>y = 1.9604 \cdot e^{0.5051x}</math> with <math>R^2 = 0.9996</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |           |           |          |   |       |       |       |   |       |       |       |   |       |       |        |   |       |       |       |   |        |        |        |   |        |        |       |



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